

Topic 8: Statistical Tests, Causal Inference and A/B Testing

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Table of contents

1	Why Statistical Tests?	2
2	Hypothesis Testing Framework	2
2.1	Interpreting p-Values	2
2.2	Type I and Type II Errors	3
3	The t-Test	3
3.1	Two-Sample Independent t-Test	3
4	The Chi-Square Test	5
5	ANOVA (Analysis of Variance)	6
6	Normality Tests	6
7	Correlation vs. Causation	7
8	Randomized Controlled Trials (RCTs)	7
9	A/B Testing	7
9.1	Typical A/B Test Workflow	7
9.2	Sample Size and Power	8
9.3	Proportions: z-Test	8
10	Summary	10

1 Why Statistical Tests?

Predictive models answer “what will happen next?” Statistical tests answer a different but equally important question: **“Is the difference I observed real, or could it be random noise?”**

Business decisions constantly depend on this question:

- Did our new website design improve conversion, or is the 2% lift within noise?
- Is the churn rate higher in region A than region B?
- Does the new training program actually raise employee productivity?

Without statistical tests, it is easy to fool yourself with random variation and make decisions that do not replicate.

Key Takeaway — The Big Idea

A statistical test helps you decide whether an observed pattern (a difference, a relationship, a deviation from a baseline) is likely due to chance or reflects a real effect. It quantifies your uncertainty so you can act with appropriate confidence.

2 Hypothesis Testing Framework

Every statistical test follows the same five-step recipe:

Key Takeaway — The Hypothesis Testing Recipe

1. **State hypotheses:**
 - H_0 (null): the status quo / “no effect”
 - H_1 (alternative): the effect you want to detect
2. **Choose a test** appropriate for the data type and question
3. **Compute the test statistic** from your sample
4. **Compute the p-value:** the probability of seeing a statistic at least this extreme if H_0 were true
5. **Decide:** if $p < \alpha$ (typically 0.05), reject H_0 ; otherwise, fail to reject

2.1 Interpreting p-Values

A **p-value** is the probability of seeing data at least as extreme as what you observed, assuming the null hypothesis is true. It is NOT the probability that the null hypothesis is true.

- **Small p-value** ($p < 0.05$): The data would be unlikely under $H_0 \Rightarrow$ reject H_0
- **Large p-value** ($p \geq 0.05$): The data is consistent with $H_0 \Rightarrow$ fail to reject

Common Mistake — What a p-Value is NOT

- NOT the probability that H_0 is true
- NOT the probability that the result happened “by chance”
- NOT a measure of the effect size (a tiny effect can have a tiny p-value with a huge sample)
- NOT a replacement for business judgment (statistical significance $\neq$ practical significance)

2.2 Type I and Type II Errors

	H_0 is True	H_0 is False
Fail to Reject H_0	Type I Error (False Positive) α	Correct (True Positive) Power = $1 - \beta$
Reject H_0	Correct (True Negative)	Type II Error (False Negative) β

- α (**significance level**): probability of a Type I error (typically 0.05)
- β : probability of a Type II error
- **Power** ($1 - \beta$): probability of correctly detecting a real effect (typically aim for 0.80)

3 The t-Test

The **t-test** compares means between groups. There are three flavors:

Test	When to use
One-sample	Compare a sample mean to a known value
Independent samples	Compare means between two separate groups
Paired	Compare two measurements on the same subjects (e.g., before/after)

3.1 Two-Sample Independent t-Test

Hypotheses:

$$H_0 : \mu_1 = \mu_2 \quad H_1 : \mu_1 \neq \mu_2$$

Test statistic (assuming equal variances):

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

where s_p is the pooled standard deviation:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

3.1.1 Worked Example

A company tests whether Campaign A generates more revenue per customer than Campaign B.

- Campaign A: $n_1 = 50$, $\bar{x}_1 = \$120$, $s_1 = \$25$
- Campaign B: $n_2 = 50$, $\bar{x}_2 = \$110$, $s_2 = \$30$

Pooled std:

$$s_p = \sqrt{\frac{49(25^2) + 49(30^2)}{98}} = \sqrt{\frac{30625 + 44100}{98}} = \sqrt{762.5} \approx 27.61$$

Test statistic:

$$t = \frac{120 - 110}{27.61 \sqrt{\frac{1}{50} + \frac{1}{50}}} = \frac{10}{27.61 \times 0.2} = \frac{10}{5.522} \approx 1.811$$

With $n_1 + n_2 - 2 = 98$ degrees of freedom, the two-sided p-value is approximately **0.073**. Since $p > 0.05$, we **fail to reject** H_0 — the observed \$10 difference is not strong enough evidence of a real effect at the 5% level.

```
from scipy import stats

t_stat, p_value = stats.ttest_ind(group_a, group_b)
print(f"t = {t_stat:.4f}, p = {p_value:.4f}")
```

4 The Chi-Square Test

The **chi-square test** checks whether two categorical variables are related. Data goes in a **contingency table** and the test compares observed frequencies to expected frequencies under independence.

Test statistic:

$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

where O_{ij} is the observed count and E_{ij} is the expected count assuming independence:

$$E_{ij} = \frac{(\text{row}_i \text{ total}) \times (\text{col}_j \text{ total})}{\text{grand total}}$$

4.0.1 Worked Example: Gender and Churn

	Churn Yes	Churn No	Row total
Female	280	720	1000
Male	260	740	1000
Col total	540	1460	2000

Expected counts (under independence):

- $E(\text{F, Yes}) = 1000 \times 540/2000 = 270$
- $E(\text{F, No}) = 1000 \times 1460/2000 = 730$
- $E(\text{M, Yes}) = 270, E(\text{M, No}) = 730$

Chi-square:

$$\begin{aligned} \chi^2 &= \frac{(280 - 270)^2}{270} + \frac{(720 - 730)^2}{730} + \frac{(260 - 270)^2}{270} + \frac{(740 - 730)^2}{730} \\ &= 0.370 + 0.137 + 0.370 + 0.137 = 1.015 \end{aligned}$$

With $(2 - 1)(2 - 1) = 1$ degree of freedom, the p-value ≈ 0.314 . We **fail to reject** H_0 — there is no evidence that churn rate differs by gender.

```
from scipy.stats import chi2_contingency

table = pd.crosstab(df["gender"], df["churn"])
chi2, p_value, dof, expected = chi2_contingency(table)
```

5 ANOVA (Analysis of Variance)

ANOVA generalizes the t-test to more than two groups. It asks: is there any difference between the group means?

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_k \quad H_1 : \text{at least one mean differs}$$

The test statistic is the F-statistic:

$$F = \frac{MS_{\text{between}}}{MS_{\text{within}}}$$

where MS_{between} is the variance between group means and MS_{within} is the average variance within groups. If the groups have truly different means, the between-group variance will be much larger than the within-group variance.

```
from scipy.stats import f_oneway

f_stat, p_value = f_oneway(group_a, group_b, group_c)
```

Important Note — ANOVA Tells You *That*, Not *Which*

A significant ANOVA result tells you *some* group differs, but not *which*. To identify which pairs differ, follow up with **post-hoc tests** (e.g., Tukey's HSD), adjusting for multiple comparisons.

6 Normality Tests

Many tests (including the t-test and ANOVA) assume the data is approximately normal. Two ways to check:

- **Visual:** histogram + Q-Q plot
- **Formal test:** Shapiro-Wilk (small samples) or D'Agostino-Pearson

```
from scipy.stats import shapiro, normaltest

stat, p = shapiro(data)    # small samples (<5000)
stat, p = normaltest(data) # larger samples
```

A significant p-value (e.g., < 0.05) means the data departs from normality. For very large samples these tests are overly sensitive — always combine with a visual check.

7 Correlation vs. Causation

Statistical tests can show that two variables are **associated**, but association alone does not prove one variable **causes** the other.

Common Mistake — Confusing Correlation with Causation

Example: Ice cream sales and drowning deaths are strongly positively correlated. It would be absurd to conclude that ice cream *causes* drowning. The real cause is a confounder: **hot weather**, which independently increases both.

Three reasons for correlation without causation:

1. **Confounding:** a third variable drives both
2. **Reverse causation:** Y causes X, not the other way around
3. **Coincidence:** random chance in a finite sample

8 Randomized Controlled Trials (RCTs)

The **gold standard** for establishing causation is the **randomized controlled trial**:

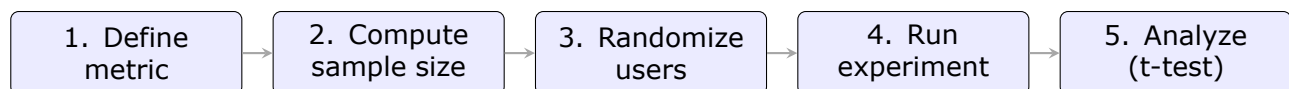
1. **Randomly assign** subjects to a treatment group and a control group
2. Apply the treatment to the treatment group, nothing (or a placebo) to control
3. Measure the outcome in both groups
4. Compare means (often with a t-test)

Random assignment ensures that — on average — the two groups are identical in every other way, so any observed difference must be caused by the treatment.

9 A/B Testing

An **A/B test** is an RCT applied to products, websites, or marketing campaigns. Users are randomly split into two (or more) groups, each sees a different version, and the team compares metrics like conversion rate, revenue, or engagement.

9.1 Typical A/B Test Workflow



9.2 Sample Size and Power

Before running an A/B test you must compute the required sample size. The formula depends on:

- α : significance level (typically 0.05)
- $1 - \beta$: desired power (typically 0.80)
- **Effect size**: smallest difference you care to detect
- **Baseline variability**: standard deviation or baseline rate

A simple formula for comparing two proportions:

$$n \approx \frac{2 \cdot (z_{\alpha/2} + z_{\beta})^2 \cdot \bar{p}(1 - \bar{p})}{\delta^2}$$

where $\bar{p}$ is the average expected proportion and δ is the minimum detectable effect.

```
from statsmodels.stats.power import NormalIndPower
from statsmodels.stats.proportion import proportion_effectsize

effect = proportion_effectsize(0.12, 0.10)    # 10% → 12% lift
analysis = NormalIndPower()
n = analysis.solve_power(effect, alpha=0.05, power=0.80)
print(f"Needed sample size per group: {n:.0f}")
```

Key Takeaway — Common A/B Testing Pitfalls

1. **Peeking**: Checking results daily and stopping when you see significance inflates Type I error
2. **Multiple testing**: Running 20 A/B tests means 1 will look significant by chance at $\alpha = 0.05$
3. **Underpowered tests**: Small samples miss real effects (high β)
4. **Novelty effects**: Users may react to any change, then revert
5. **Ignoring business significance**: A 0.1% lift with $p < 0.001$ may not be worth shipping

9.3 Proportions: z-Test

For binary outcomes (converted or not), the A/B test is usually a **two-proportion z-test**:

$$z = \frac{\hat{p}_A - \hat{p}_B}{\sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_A} + \frac{1}{n_B} \right)}}$$

where $\hat{p}$ is the pooled proportion.

9.3.1 Worked Example

A landing page A/B test:

- Control (A): 10,000 visitors, 520 conversions ($\hat{p}_A = 0.052$, 5.2%)
- Variant (B): 10,000 visitors, 600 conversions ($\hat{p}_B = 0.060$, 6.0%)

Pooled proportion: $\hat{p} = (520 + 600)/20,000 = 0.056$.

$$z = \frac{0.060 - 0.052}{\sqrt{0.056 \times 0.944 \times \left(\frac{1}{10,000} + \frac{1}{10,000}\right)}}$$

$$= \frac{0.008}{\sqrt{0.0528 \times 0.0002}} = \frac{0.008}{\sqrt{0.00001056}} = \frac{0.008}{0.003249} \approx 2.462$$

Two-sided p-value $\approx 0.014 < 0.05$. Reject H_0 . The 0.8 percentage point lift in conversion is statistically significant. Whether to ship depends on the business context — the sample was large, so even a small effect reached significance, and the team should weigh the practical value.

Practice — A/B Test Interpretation

A product team runs an A/B test comparing the old checkout flow (A) with a redesigned one (B). Results:

- Group A: 5,000 users, 350 purchases (7.0%)
- Group B: 5,000 users, 410 purchases (8.2%)

- What are H_0 and H_1 ?
- Compute the pooled proportion and the z-statistic.
- Is the difference significant at $\alpha = 0.05$?
- The team also noticed that B took 0.8 seconds longer to load. Does that change your interpretation?

Answer

- (a) $H_0 : p_A = p_B$ (no difference); $H_1 : p_A \neq p_B$.
- (b) Pooled: $\hat{p} = (350 + 410)/10,000 = 0.076$. Standard error: $\sqrt{0.076 \times 0.924 \times (2/5000)} = \sqrt{0.0000281} \approx 0.005298$. $z = (0.082 - 0.070)/0.005298 \approx 2.265$. Two-sided p-value ≈ 0.024 .
- (c) Yes, $p < 0.05$ — reject H_0 . The new flow has a statistically significant higher conversion rate.
- (d) The load-time regression is a confounder. If B is slower, real users might abandon more on slow connections, shrinking the effect in production. Also consider whether slower load time has knock-on effects (user frustration, retention). Statistical significance alone is not sufficient; weigh total business impact.

10 Summary

What We Covered

Topic	Key Concepts
Hypothesis testing	H_0 vs H_1 ; p-values; Type I & II errors
t-Test	Compares means; one-sample, independent, paired
Chi-Square	Tests independence of categorical variables
ANOVA	Compares means across 3+ groups
Normality tests	Shapiro-Wilk, D'Agostino-Pearson
Correlation vs. causation	RCTs are the gold standard
A/B testing	Randomized experiment for products
Sample size & power	Plan before running; avoid peeking

Next week: Buffer/flex week — catch-up, case study, or review before the final exam.